Final Report: Does Money Buy Points in Football?

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Executive summary

Squad market value explains 57% of the differences in points between clubs in Europe's top five leagues; the remaining 43% depends on other factors, and that is where a mid-budget club can compete.

- **Money matters, but it doesn't decide everything.** Doubling squad value is associated with about 11.5 more points per season.
- **The weight of money varies by league.** It explains 71% of points in Serie A, but only 49% in the Bundesliga.
- **Some modest clubs consistently beat the odds.** 15 clubs with squads below their league average earn more points than expected; RC Lens, Union Berlin and SD Eibar are the clearest cases.

Interactive dashboard: Does money buy points in football? (Tableau Public) (labels in Spanish)

Context: capstone case study for the Google Data Analytics Professional Certificate. Tools: BigQuery (SQL), Google Sheets and Tableau Public.

1. Ask: the business task

The client is a mid-budget football club in one of Europe's top five leagues that wants to compete better without spending like the big clubs.

Business task statement (deliverable 1):

> Analyze the relationship between squad market value and league points in Europe's top five leagues from 2015/16 to 2024/25, to measure how strong that relationship is and identify which clubs earn more points than their investment predicts, in order to recommend more efficient squad-management practices to the client.

Stakeholders: sporting director (decides signings), CFO (approves the budget) and club president. They are business people, not technical: they need a few clear charts and concrete recommendations.

Metrics:

Metric	Definition	Why
Squad value	Sum of the market value of every player who played for the club that season, taken at September 1 (€)	Measures squad investment
Relative value	Squad value ÷ league average that season	Removes market inflation and differences between leagues
Points per game (PPG)	(3 × wins + draws) ÷ games	Compares 34- and 38-game leagues and incomplete seasons
Expected PPG	Prediction of a linear regression of PPG on ln(relative value)	Benchmark of what each club "should" earn
Points above expected	(actual PPG − expected PPG) × games	Positive = efficient club
R^2	Share of the variation in PPG explained by relative value	Measures how much money decides

2. Prepare: data sources

The only source is the public Football Data from Transfermarkt dataset on Kaggle, built from Transfermarkt and updated weekly (code on GitHub).

Tables used (deliverable 2):

Table	Rows	Content	Use in the analysis
appearances	1,894,350	Every player appearance: club, game, minutes	Identify each season's squad
Player_valuations	656,301	Market value history per player, with date	Calculate squad value

Table	Rows	Content	Use in the analysis
club_games	177,916	Each game from each club's side: goals for and against	Calculate points
Transfers	175,165	Transfers between clubs	Not used in this version
Games	88,958	One game per row: league, season, result	Filter leagues and seasons
clubs	796	Name and league of each club	Club names in results

Scope: Premier League (GB1), LaLiga (ES1), Serie A (IT1), Bundesliga (L1) and Ligue 1 (FR1), seasons 2015/16 to 2024/25. The current season is excluded because it is incomplete.

ROCCC assessment:

Criterion	Level	Comment
Reliable	Medium	Transfermarkt is the most widely used reference, but its values are estimates, not real prices
Original	Low	Secondary source: data scraped from Transfermarkt, not from the clubs
Comprehensive	High	Covers every game and club in scope
Current	High	Updated weekly
Cited	High	Open, documented project widely used in sports analytics

Limitations:

- Market value is not the real wage bill; salaries are rarely public.
- A club may look efficient because Transfermarkt undervalues its players, not only because of good management.
- The data contains only public statistics on professional players, with no sensitive personal data.

3. Process: cleaning and transformation

The data was loaded and cleaned in BigQuery into a final table of 976 rows (one per club and season), with no rows lost in the joins. Every step is recorded in the cleaning log (deliverable 3, in Spanish).

Tools: BigQuery to join multi-million-row tables with SQL, Google Sheets for the cleaning log and Tableau Public for the visualizations.

Main steps:

1. **Load.** The CSV files were uploaded to the `Caso_de_estudio` dataset. The `appearances` table exceeded BigQuery's 100 MB local upload limit, so it was loaded from Google Drive.
2. **Load check.** Row counts were verified for every table. `club_games` has exactly twice the rows of `Games` (177,916 = 2 × 88,958), as expected, since each game appears once per club.
3. **Data types.** Dates were confirmed as DATE, and goals and values as integers.
4. **Filtering.** Only the 5 leagues and seasons 2015 to 2024; games without recorded goals were excluded.
5. **Intermediate tables.** `puntos` (points and goals per club-season) and `valor_plantilla` (each player's latest valuation before September 1, for every player who appeared for the club).
6. **Final table `club_temporada`.** Joins both tables with `clubs` and adds PPG and relative value.

Integrity check: all three tables have 976 rows, which match reality exactly: 200 club-seasons in the Premier League, LaLiga and Serie A (20 teams × 10), 180 in the Bundesliga (18 × 10) and 196 in Ligue 1 (20 teams × 8 seasons + 18 × 2 since 2023). No squad has fewer than 20 valued players, so no club was excluded.

4. Analyze: analysis summary

The correlation between relative value and points per game is 0.75 ($R^2 = 0.571$, $p < 0.0001$): a strong relationship that still leaves almost half of the variation unexplained (deliverable 4).

Descriptive statistics by league:

League	Club-seasons	Average value (€M)	Minimum (€M)	Maximum (€M)
Premier League	200	417.5	67.3	1,318.5
LaLiga	200	245.9	30.8	1,340.5
Serie A	200	224.0	23.0	824.2
Bundesliga	180	215.2	18.9	964.2
Ligue 1	196	157.0	13.9	1,077.5

The average Premier League squad is worth 2.7 times the average Ligue 1 squad, and in Ligue 1 the most expensive club is worth about 7 times the league average. That is why the analysis uses relative value instead of euros.

Correlation by league, with and without logarithm:

League	Plain correlation	Correlation with ln	R^2 with ln
Serie A	0.83	0.84	71%
LaLiga	0.81	0.78	61%
Premier League	0.76	0.75	56%
Ligue 1	0.72	0.75	56%
Bundesliga	0.74	0.70	49%

The two versions differ by 0.04 or less. The logarithm was chosen because it reduces the distortion of extreme clubs such as PSG: going from 50 to 100 million changes a team far more than going from 1,000 to 1,050.

Expected points model (simple linear regression computed in SQL and validated with Tableau's trend line):

```
\text{Expected PPG} = 0.4386 \cdot \ln(\text{relative value}) + 1.518
```

Doubling relative value adds $0.4386 \times \ln 2 \approx 0.30$ points per game, about 11.5 points over a 38-game season.

Efficiency ranking: points above expected were averaged for the 96 clubs with 5 or more seasons in the period, then filtered to those with an average relative value below 1.

5. Share: visualizations and key findings

The five visualizations are combined in the Tableau Public dashboard; each one supports one finding (deliverable 5).

#	Visualization	Finding
1	Scatter plot: relative value (log axis) vs. points per game, with trend line	Money explains 57% of the differences in points, but not everything
2	Bars: % of points explained by money in each league	Money matters most in Serie A (71%) and least in the Bundesliga (49%)
3	Lines: average squad value by league, 2015–2024	The Premier League went from €213M to €606M; LaLiga, Serie A and the Bundesliga have stalled at €240–280M since 2019
4	Diverging bars: 10 most and 10 least efficient clubs	RC Lens earns +14.4 points per season above expected; FC Schalke 04, −11.5
5	Lines: actual vs. expected PPG for a selected club (RC Lens by default)	Lens beat expectations in all 5 seasons, with ~30 extra points in 2022, when it finished second in France

Efficient clubs on a low budget (average relative value below 1, 5 or more seasons): the closest role models for the client.

Club	League	Average relative value	Points above expected per season	Seasons above expected
RC Lens	Ligue 1	0.66	+14.4	5 of 5
1.FC Union Berlin	Bundesliga	0.42	+10.7	4 of 6
Girona FC	LaLiga	0.45	+9.2	3 of 5
SD Eibar	LaLiga	0.27	+9.1	5 of 6
Stade Brestois 29	Ligue 1	0.40	+7.5	4 of 6
CA Osasuna	LaLiga	0.35	+6.2	5 of 7

Club	League	Average relative value	Points above expected per season	Seasons above expected
SC Freiburg	Bundesliga	0.51	+6.0	6 of 9
Deportivo Alavés	LaLiga	0.31	+5.9	5 of 8
LOSC Lille	Ligue 1	0.96	+5.9	7 of 10
Rayo Vallecano	LaLiga	0.28	+5.8	3 of 6
RCD Mallorca	LaLiga	0.31	+4.1	3 of 5
RC Strasbourg Alsace	Ligue 1	0.53	+3.0	5 of 8
Villarreal CF	LaLiga	0.98	+3.0	6 of 10
CD Leganés	LaLiga	0.29	+2.9	4 of 5
Athletic Bilbao	LaLiga	0.82	+2.0	5 of 10

Nine of the 15 are from LaLiga, four from Ligue 1, two from the Bundesliga and none from the Premier League or Serie A. The most consistent are Lens, Eibar, Freiburg and Lille; Girona, Rayo and Mallorca rely more on isolated seasons.

6. Act: conclusions and recommendations

Money buys a large share of points, but a mid-budget club has real room to outperform its investment.

Top high-level insights (deliverable 7):

1. **Squad value explains 57% of the differences in points.** Doubling it is associated with about 11.5 more points per season; the other 43% depends on management, coaching and other factors.
2. **The weight of money depends on the league.** It ranges from 71% in Serie A to 49% in the Bundesliga, and the financial gap is widening: the Premier League nearly tripled its squad values since 2015 while the other leagues stalled after 2019.
3. **Sustained efficiency exists.** 15 clubs with below-average squads beat expectations; Lens, Eibar, Freiburg and Lille do so in most of their seasons.

Recommendations for the client:

1. **Use expected points as a realistic target.** Set sporting goals and evaluate the coaching staff against what the budget predicts, not against the big clubs.
2. **Study the efficient clubs' model.** Analyze how Lens, Union Berlin, Freiburg and Eibar sign, develop and sell players.
3. **Judge every signing by performance per euro.** Spending alone is no protection: Schalke 04, Southampton, Valencia, Genoa and Everton had sizeable budgets and earned between 7.5 and 11.5 fewer points than expected per season.

Further analysis recommended (deliverable 6):

- **Coaching stability.** Freiburg and Union Berlin kept the same head coach for many years; `club_games` includes the coach's name and would allow testing this hypothesis.
- **Real wages**, where public data exists, to complement market value.
- **Transfer returns** using the `Transfers` table: fee paid vs. resale value.
- **Squad age** and its relationship with efficiency.
- **Relegated clubs.** Requiring 5 top-flight seasons creates survivorship bias, because it leaves out teams that went down quickly.
- **Other leagues**, such as Liga MX, if coverage is sufficient.

Caveat: the analysis shows association, not causation. A club that performs well can also raise its players' value, not only the other way around.

Appendix: main SQL queries

These four BigQuery queries build the analysis tables (table and column names are in Spanish: `puntos` = points, `valor_plantilla` = squad value, `club_temporada` = club-season, `eficiencia` = efficiency).

Points per club and season:

```
CREATE OR REPLACE TABLE `Caso_de_estudio.puntos` AS
SELECT g.season, g.competition_id, cg.club_id,
       COUNT(*) AS partidos,
       SUM(CASE WHEN cg.own_goals > cg.opponent_goals THEN 3
                WHEN cg.own_goals = cg.opponent_goals THEN 1
                ELSE 0 END) AS puntos,
       SUM(cg.own_goals) AS goles_favor,
       SUM(cg.opponent_goals) AS goles_contra
FROM `Caso_de_estudio.club_games` cg
JOIN `Caso_de_estudio.Games` g ON g.game_id = cg.game_id
WHERE g.competition_id IN ('GB1','ES1','IT1','L1','FR1')
  AND g.season BETWEEN 2015 AND 2024
  AND cg.own_goals IS NOT NULL
GROUP BY 1, 2, 3;
```

Squad value at the start of each season:

```
CREATE OR REPLACE TABLE `Caso_de_estudio.valor_plantilla` AS
WITH plantilla AS (
  SELECT DISTINCT g.season, a.player_club_id AS club_id, a.player_id
  FROM `Caso_de_estudio.appearances_drive` a
  JOIN `Caso_de_estudio.Games` g ON g.game_id = a.game_id
  WHERE g.competition_id IN ('GB1','ES1','IT1','L1','FR1')
    AND g.season BETWEEN 2015 AND 2024
),
valor AS (
  SELECT p.season, p.club_id, p.player_id, v.market_value_in_eur
  FROM plantilla p
  JOIN `Caso_de_estudio.Player_valuations` v
    ON v.player_id = p.player_id
   AND v.date <= DATE(p.season, 9, 1)
  QUALIFY ROW_NUMBER() OVER (
    PARTITION BY p.season, p.club_id, p.player_id
    ORDER BY v.date DESC) = 1
)
SELECT season, club_id,
       SUM(market_value_in_eur) AS valor_plantilla_eur,
       COUNT(*) AS jugadores_valorados
FROM valor
GROUP BY 1, 2;
```

Final table with relative value:

```
CREATE OR REPLACE TABLE `Caso_de_estudio.club_temporada` AS
SELECT p.season, p.competition_id, p.club_id, c.name AS club,
       p.partidos, p.puntos,
       ROUND(p.puntos / p.partidos, 2) AS puntos_por_partido,
       v.valor_plantilla_eur, v.jugadores_valorados,
       ROUND(v.valor_plantilla_eur /
         AVG(v.valor_plantilla_eur) OVER (PARTITION BY p.season,
p.competition_id), 2) AS valor_relativo
FROM `Caso_de_estudio.puntos` p
JOIN `Caso_de_estudio.valor_plantilla` v USING (season, club_id)
JOIN `Caso_de_estudio.clubs` c ON c.club_id = p.club_id;
```

Expected points (simple linear regression):

```
CREATE OR REPLACE TABLE `Caso_de_estudio.eficiencia` AS
WITH base AS (
  SELECT *, LN(valor_relativo) AS x
  FROM `Caso_de_estudio.club_temporada`
),
modelo AS (
  SELECT
    COVAR_SAMP(x, puntos_por_partido) / VAR_SAMP(x) AS pendiente,
    AVG(puntos_por_partido) - COVAR_SAMP(x, puntos_por_partido) / VAR_SAMP(x)
* AVG(x) AS intercepto
  FROM base
)
SELECT b.season, b.competition_id, b.club, b.partidos, b.puntos,
       b.puntos_por_partido, b.valor_plantilla_eur, b.valor_relativo,
       ROUND(m.intercepto + m.pendiente * b.x, 2) AS ppp_esperado,
       ROUND(b.puntos_por_partido - (m.intercepto + m.pendiente * b.x), 2) AS
ppp_sobre_esperado,
       ROUND((b.puntos_por_partido - (m.intercepto + m.pendiente * b.x)) *
b.partidos, 1) AS puntos_sobre_esperado
FROM base b CROSS JOIN modelo m;
```

Sources

- Football Data from Transfermarkt (Kaggle)
- transfermarkt-datasets (GitHub)
- Dashboard on Tableau Public

- Cleaning log (Google Sheets, in Spanish)